

Feature	Fires on	Energy	Steering vs. random	Verdict
11326	Basic (R/K) + Pro, before acidic/Pro-rich	−12.9 (act, q=0.005)	+0.8pp (p=0.21, n=1680)	energy only
233	Basic/acidic after an LSE/LSEE motif	−13.5 (act, q=0.011)	+2.8pp (p<0.001, n=2184)	energy + causal steering
6073	Leu preceded by acidic (D/E)	−30.1 (pres, q=0.011)	+1.2pp (p=1.00, n=84)	energy only
10586	Hydrophobic (G/L/F/I/W); iGluR-associated	−26.6 (pres, q=0.059)	−1.3pp (p=1.00, n=78)	no detected effect
14247	Hydrophobic (I/L/F); no dominant domain	−11.3 (act, q=0.064)	+1.4pp (p=0.01, n=2865)	no detected effect
4657	Ser after acidic; RTN1-4/CAPS family	−22.1 (pres, q=0.064)	−1.1pp (p=0.51, n=261)	no detected effect
12588	MUN domain (IPR010439)	−37.8 (pres, q=0.329)	+3.3pp (p=0.50, n=60)	no detected effect
12918	Phe in TGLQG[F]L; tubulin/FtsZ GTPase	−8.7 (act, q=0.532)	+2.8pp (p=0.75, n=72)	no detected effect
1707	No clear pattern	+25.0 (pres, q=0.695)	+0.7pp (p=0.15, n=867)	no detected effect
2214	Glycoside hydrolase superfamily	+13.0 (pres, q=0.901)	+5.0pp (p=0.58, n=60)	no detected effect
6869	Gly before Lys (G-K); P-loop NTPase/GTPase	+3.7 (act, q=0.987)	+1.5pp (p=1.00, n=66)	no detected effect

Energy: confound-controlled regression, n=477 designs, BH-FDR. Steering: paired vs. norm-matched random, ≤2×, Bonferroni threshold p<0.0045.